Kali VM Setup

After installing your Kali VM, this guide will get you connected to our Pentest Lab.

1. Run the following command to update and upgrade your Kali packages

```
sudo apt update && sudo apt full-upgrade -y
```

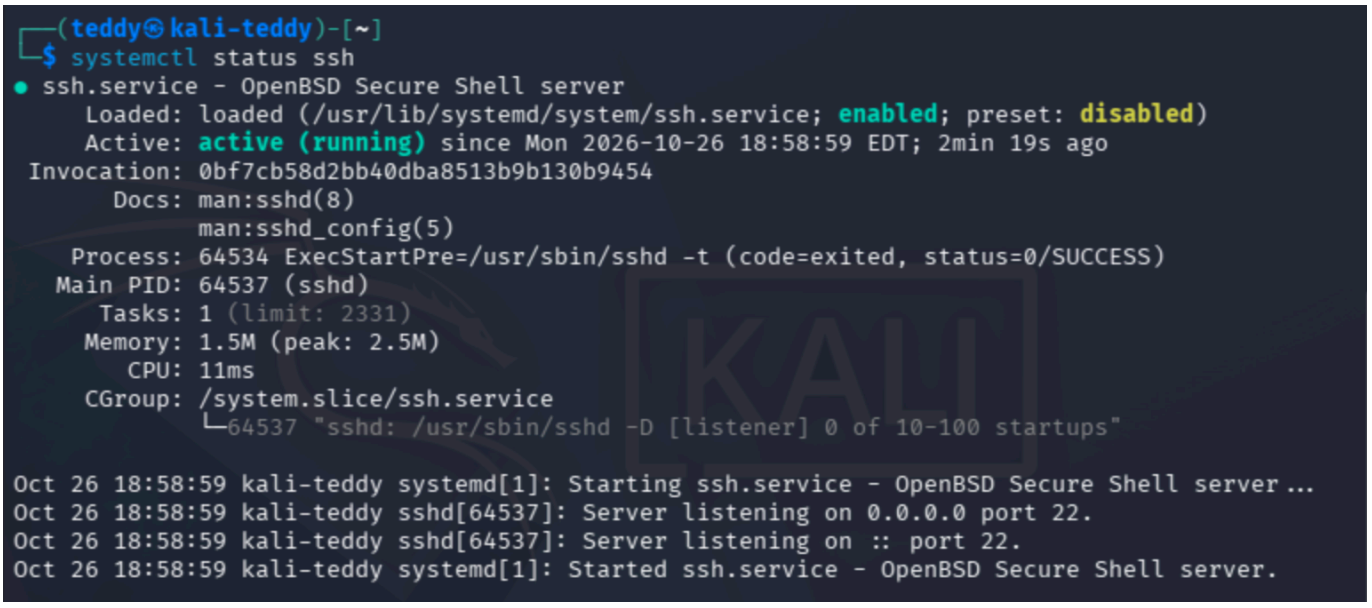
2. Enable `ssh` so we can easily transfer files from your machine to kali VM

```
sudo systemctl enable --now ssh
```

3. Verify `ssh` is `enabled` and `active`

```
systemctl status ssh
```

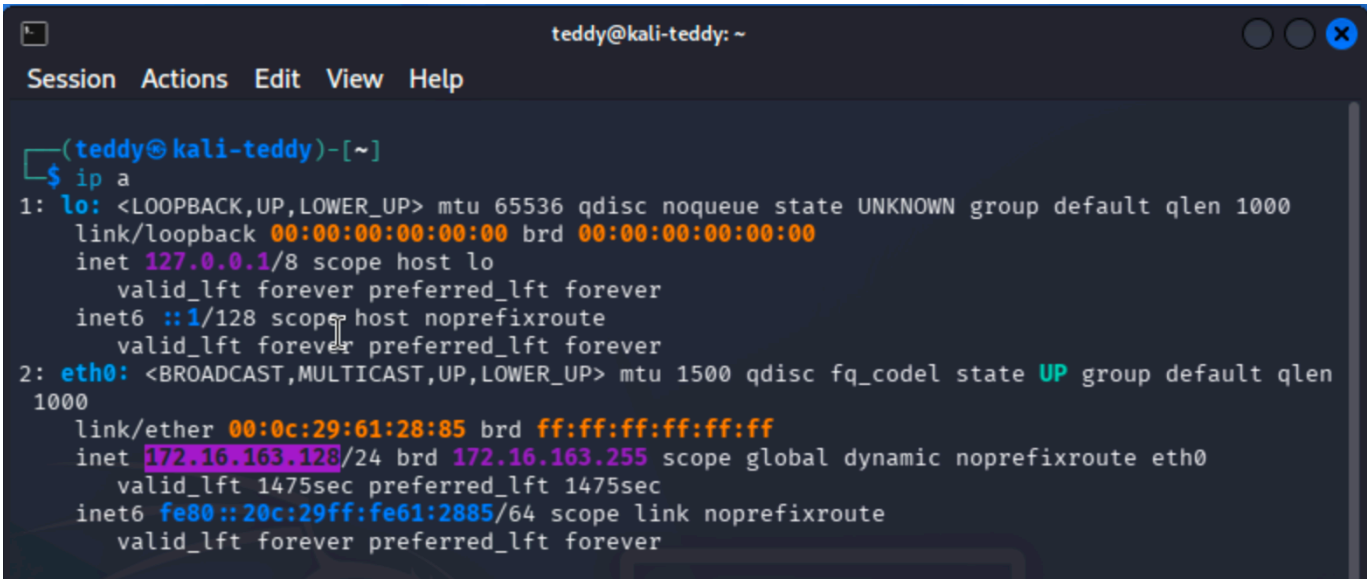
You should see something like the following output:



4. Locate the IP address of your Kali VM

```
ip a
```

Look for the device `eth0` and the `inet` reference:



> My IP address is `172.16.163.128`

5. `ssh` into your Kali VM with `user@ip-address` from a terminal on your local machine

```
ssh teddy@172.16.163.128
```

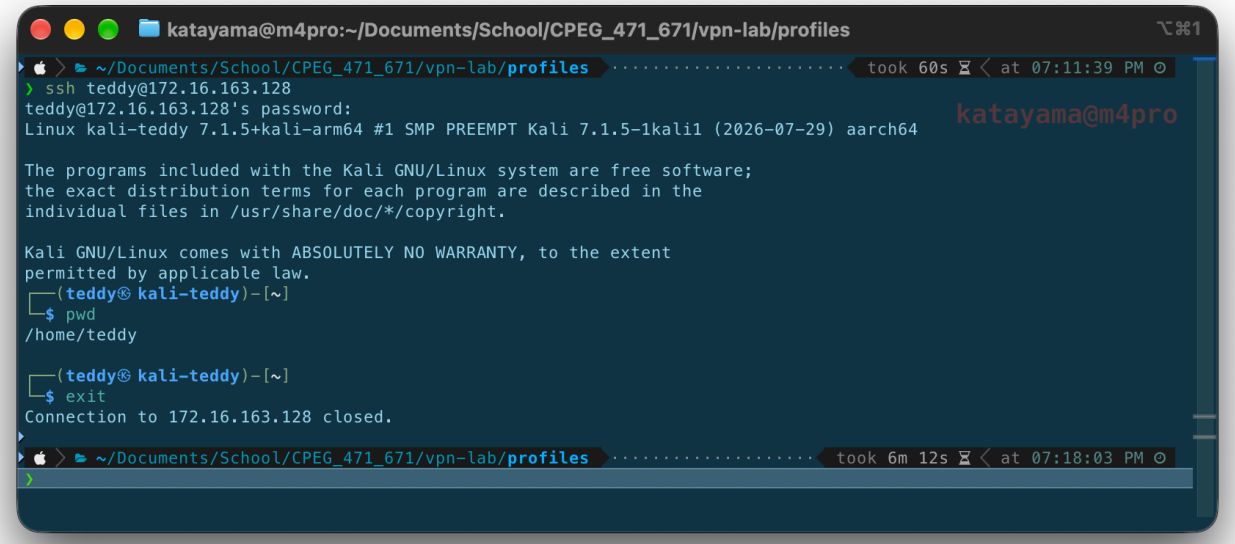
Run `pwd` to "print working directory" of your "home"

```
pwd
```

My home directory is `/home/teddy`

Then run `exit` to return to your local machine.

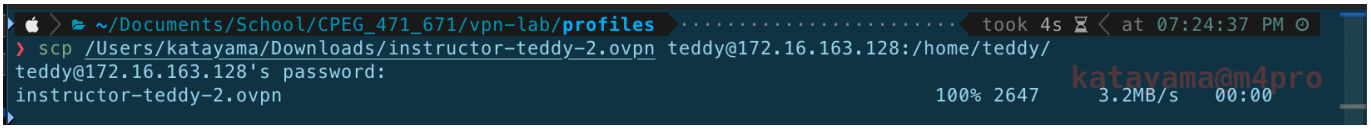
```
exit
```



6. Now transfer the OVPN config file you received via email using `scp` command

> I downloaded mine to `/Users/katayama/Downloads/instructor-teddy-2.ovpn`

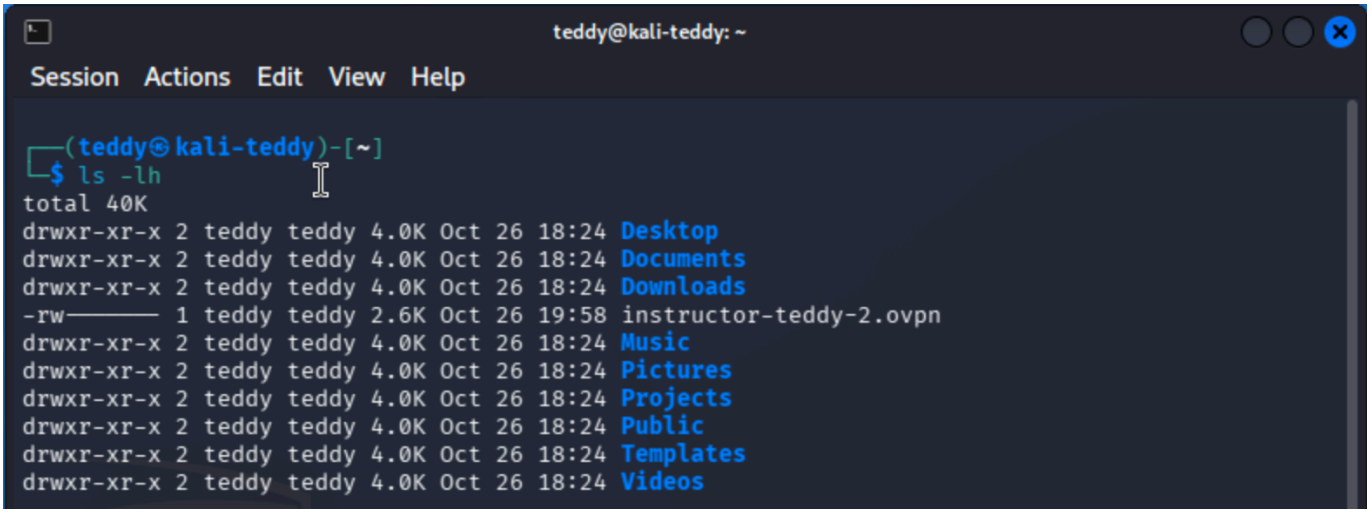
```
scp /Users/katayama/Downloads/instructor-teddy-2.ovpn
teddy@172.16.163.128:/home/teddy/.
```



> With the transfer completed, we can return to our Kali VM and configure it for persistent Open VPN access to our Pentest Lab

8. In a terminal on your Kali VM, verify the ovpn file successfully transferred with `ls` command:

```
ls -lh
```



9. Move your `ovpn` file to `/etc/openvpn/openvpn.conf`

```
sudo mv instructor-teddy-2.ovpn /etc/openvpn/openvpn.conf
```

10. Confirm the `ovpn` has been moved:

```
ls -lh /etc/openvpn
```

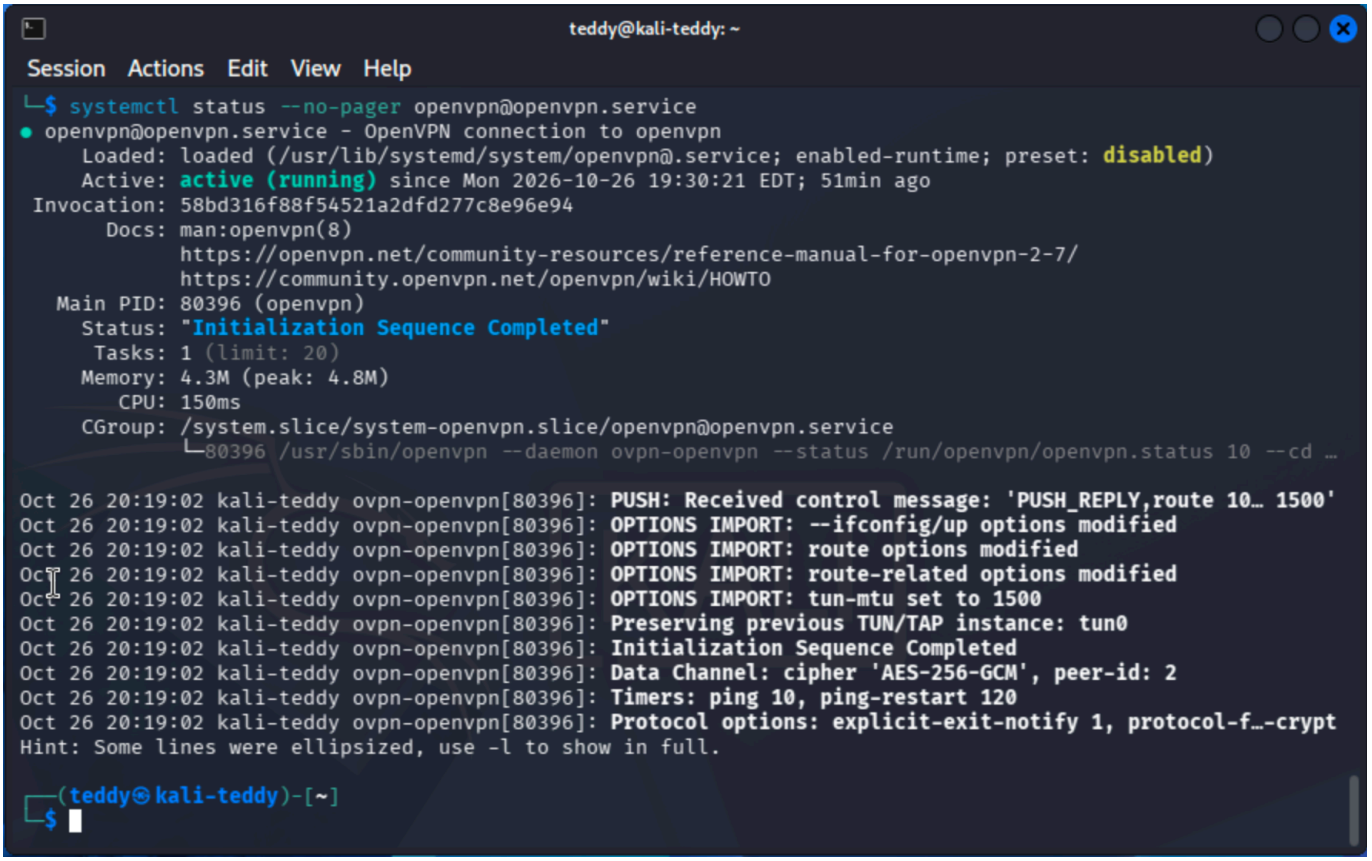
```
$ ls -lh /etc/openvpn
total 16K
drwxr-xr-x 2 root  root  4.0K Jul  1 15:46 client
-rw------- 1 teddy teddy 2.6K Oct 26 19:24 openvpn.conf
drwxr-xr-x 2 root  root  4.0K Jul  1 15:46 server
-rwxr-xr-x 1 root  root  1.5K Jul  1 15:46 update-resolv-conf
```

11. Enable and start the OpenVPN service

```
sudo systemctl enable --now openvpn
```

12. Check the OpenVPN service is `active` and `running`

```
systemctl status --no-pager openvpn@openvpn.service
```



13. Now run the `ip a` command, you should see a new tunnel device `tun0` has been added

```
ip a
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group
default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
group default qlen 1000
    link/ether 00:0c:29:61:28:85 brd ff:ff:ff:ff:ff:ff
    inet 172.16.163.128/24 brd 172.16.163.255 scope global dynamic noprefixroute
eth0
       valid_lft 996sec preferred_lft 996sec
    inet6 fe80::20c:29ff:fe61:2885/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
5: tun0: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group
default qlen 1000
    link/none
    inet 10.8.0.21/24 brd 10.8.0.255 scope global tun0
       valid_lft forever preferred_lft forever
    inet6 fe80::9303:3c80:5401:f562/64 scope link stable-privacy proto kernel_ll
       valid_lft forever preferred_lft forever
```

> Take note of your `tun0` IP address and submit it to canvas, mine is `10.8.0.21`